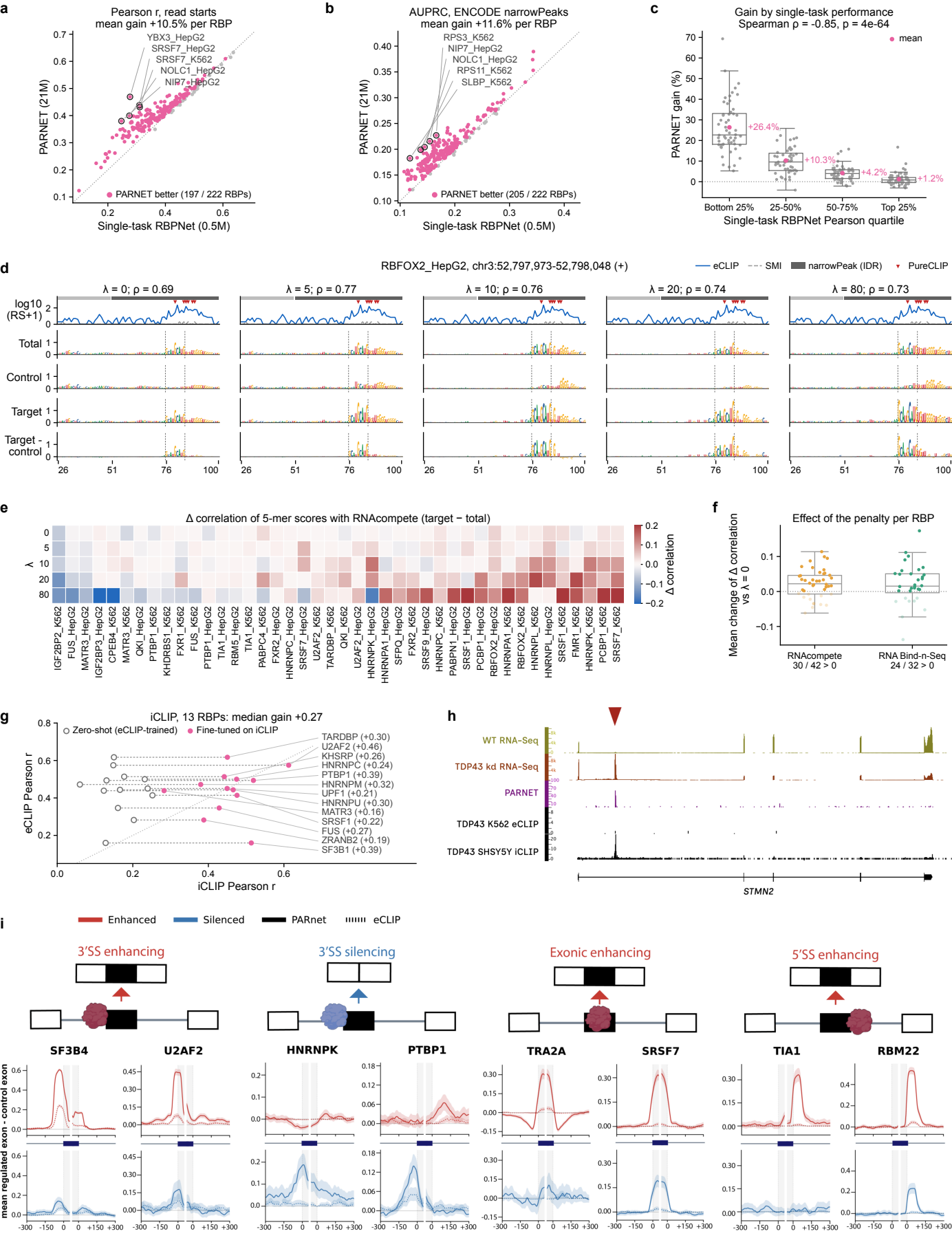

a
Pearson r, read starts
mean gain +10.5% per RBP
YBX3_HepG2
SRSF7_HepG2
SRSF7_K562
NOLC1_HepG2
NIP7_HepG2
PARNET (21M)
PARNET better (197 / 222 RBPs)
Single-task RBPNet (0.5M)
b
AUPRC, ENCODE narrowPeaks
mean gain +11.6% per RBP
RPS3_K562
NIP7_HepG2
NOLC1_HepG2
RPS11_K562
SLBP_K562
PARNET (21M)
PARNET better (205 / 222 RBPs)
Single-task RBPNet (0.5M)
c
Gain by single-task performance
Spearman ρ = -0.85, p = 4e-64
mean
PARNET gain (%)
+26.4%
+10.3%
+4.2%
+1.2%
Bottom 25%
25-50%
50-75%
Top 25%
Single-task RBPNet Pearson quartile
d
RBFOX2_HepG2, chr3:52,797,973-52,798,048 (+)
eCLIP
SMI
narrowPeak (IDR)
PureCLIP
λ = 0; ρ = 0.69
λ = 5; ρ = 0.77
λ = 10; ρ = 0.76
λ = 20; ρ = 0.74
λ = 80; ρ = 0.73
log10 (RS+1)
Total
Control
Target
Target - control
e
Δ correlation of 5-mer scores with RNAcompete (target − total)
λ
Δ correlation
IGF2BP2_K562
FUS_HepG2
MATR3_HepG2
IGF2BP3_HepG2
CPEB4_K562
MATR3_K562
QKI_HepG2
PTBP1_K562
KHDRBS1_K562
FXR1_K562
FUS_K562
PTBP1_HepG2
TIA1_HepG2
RBM5_HepG2
TIA1_K562
PABPC4_K562
FXR2_HepG2
HNRNPC_HepG2
SRSF7_HepG2
U2AF2_K562
TARDBP_K562
QKI_K562
U2AF2_HepG2
HNRNPK_HepG2
HNRNPA1_HepG2
SFPQ_HepG2
FXR2_K562
SRSF9_HepG2
HNRNPC_K562
PABPN1_HepG2
SRSF1_HepG2
PCBP1_HepG2
RBFOX2_HepG2
HNRNPA1_K562
RBFOX2_K562
HNRNPL_K562
HNRNPL_HepG2
SRSF1_K562
FMR1_K562
HNRNPK_K562
PCBP1_K562
SRSF7_K562
f
Effect of the penalty per RBP
Mean change of Δ correlation vs λ = 0
RNAcompete
30 / 42 > 0
RNA Bind-n-Seq
24 / 32 > 0
g
iCLIP, 13 RBPs: median gain +0.27
Zero-shot (eCLIP-trained)
Fine-tuned on iCLIP
eCLIP Pearson r
iCLIP Pearson r
TARDBP (+0.30)
U2AF2 (+0.46)
KHSRP (+0.26)
HNRNPC (+0.24)
PTBP1 (+0.39)
HNRNPM (+0.32)
UPF1 (+0.21)
HNRNPU (+0.30)
MATR3 (+0.16)
SRSF1 (+0.22)
FUS (+0.27)
ZRANB2 (+0.19)
SF3B1 (+0.39)
h
WT RNA-Seq
TDP43 kd RNA-Seq
PARNET
TDP43 K562 eCLIP
TDP43 SHSY5Y iCLIP
STMN2
i
Enhanced
Silenced
PARnet
eCLIP
3'SS enhancing
3'SS silencing
Exonic enhancing
5'SS enhancing
SF3B4
U2AF2
HNRNPK
PTBP1
TRA2A
SRSF7
TIA1
RBM22
mean regulated exon - control exon